Lab 10 - Filament Diagnostics - Diameter, Temperature and Flow

Course: Innovating Product Development with 3D Printing and Additive Manufacturing (TGS-2022017597) Topic 02: Process Related Issues for AM TSC objective: A2, A3 - analyse material-side causes of extrusion issues and verify filament quality

Goal

Measure filament diameter with digital calipers, verify it against the slicer's material setting, and run a temperature check to place your filament in its working window.

What you'll build

A verified filament profile plus retraction and cooling evidence from purpose-built diagnostic models.

Tools: Digital calipers, filament spool (PLA/ABS/PETG), Ultimaker Cura, retraction-test.stl, cooling-spire.stl, FDM printer

Steps

1. Cut off about 30 cm of filament from the spool end.

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Flush cutters - ~30 cm sample
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2. Measure the diameter with digital calipers at five points along the sample, rotating the filament at each point.

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Calipers - 5 readings, 2 orientations each
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3. Average the readings; a good 1.75 mm filament stays within ±0.05 mm.

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Expect 1.70-1.80 mm
```

4. Enter the measured diameter in Cura's material settings so flow is computed from reality, not the label.

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Print Settings > Material > Diameter - measured value
```

5. Check the material's temperature window: PLA 195-215 °C, PETG 240-260 °C, ABS 240-260 °C, Nylon 240-260 °C.

```
Material box / temperature table
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6. Set the printing temperature to the middle of the window and print a small test cube; drop 5 °C if strings appear, raise 5 °C if layers don't bond.

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Material > Printing Temperature - tune in 5  degrees C steps
```

7. Watch the first layers: clicking from the extruder or ground filament means the temperature is too low or the nozzle is partially blocked.

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Listen: extruder clicks = under-extrusion warning
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8. Slice retraction-test.stl with your baseline retraction, then change only one variable at a time (distance or speed); compare strings without masking jams or surface dimples.

```
Retraction > record distance + speed > Preview > optional test print
```

9. Slice cooling-spire.stl or twist-with-cooling-tower.stl and compare the upper layers with minimum layer time/fan control enabled and disabled.

```
Cooling > Enable Print Cooling > Minimum Layer Time > compare tip quality
```

Evidence to capture

- Five filament-diameter readings and the calculated average.
- Retraction matrix recording distance, speed, stringing and surface defects.
- Cooling comparison recording fan/minimum-layer-time settings and tip quality.

Test it

Your diameter average is within ±0.05 mm of nominal, Cura carries the measured value, and the diagnostic models support a retraction/cooling choice with less stringing and a cleaner spire without introducing gaps or jams.

Files in this folder

- retraction-test.stl - Multi-tower benchmark that exposes stringing during travel moves.
- cooling-spire.stl - Tall narrowing spire that reveals insufficient layer cooling.
- twist-with-cooling-tower.stl - Combined geometry for observing minimum-layer-time and cooling behaviour.